QUANT

Number System

(Digital Root, Unit's Digit, Last two digits
& other important concepts)

Assignment 03

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Numbers Online Assignment -03

1. $P^3 + 2P$, for any positive integer P, is always divisible by

1. 4 2. 3 3. 5 4. 0

2. The sum of first P natural numbers is Q. If the next number is also added the sum becomes $2Q$. Find the value of $(P + Q)^2$.

1. 49 2. 5 3. 25 4. None of these

3. $G(x)$ = Sum of all the digits of natural number x. e.g. G(182) = 1 + 8 + 2 = 11. What is the value of P? where P = {G (251) + G (252) + G (253) + …. + G (300)}

1. 676 2. 670 3. 671 4. 666

4. For how many values of a are, a, $a + 14$, $a + 26$ prime numbers?

1. One 2. Two 3. None 4. Infinite

5. The sum of four prime numbers is 43. Which of the following could be the product of these prime numbers?

1. 1,155 2. 4,862 3. 1,495 4. 2,719

6. The sum of the first K (24 > K > 20) natural numbers is always divisible by (it being given that K is even.)

1. K + 1 2. 11 3. 23 4. All of these

DIRECTIONS for questions 7– 8: *If the last two digits of a five-digit number are interchanged, the new number obtained is greater than the original number by 36.*

7. What is the sum of the last two digits of the number?

1. 6 2. 14 3. 9 4. Data inadequate

8. What is the difference between the last two digits?

1. 6 2. 4 3. 5 4. Data inadequate

9. Let x, y and z be distinct integers, that are odd and positive. Which one of the following statements cannot be true?

1. $xy \times z^2$ is odd
2. $[(x - y)^2]\,z$ is even
3. $(x + y - z)^2 \times (x + y)$ is even
4. $(x - y)(y + 2)(x + y - z)$ is odd

10. Given x and n are integers, $\frac{15n^3 + 6n^2 + 5n + x}{n}$ is not an integer for what condition?

1. n is positive
2. x is divisible by n
3. x is not divisible by n
4. Both 1 & 3 options

Answer keys & Explanations

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1.	2	Take P = 1 and find the value as 3. Then take P as 2 and get the value to be 12. As both the numbers are divisible by 3, thus 3, given in the 2nd option is the answer.
2.	3	The sum of first P natural numbers will be $\frac{P(P+1)}{2}$ and the sum of first (P + 1) numbers will be (P + 1) $\frac{(P+2)}{2}$. As per the statement of the question $(P + 1)(P + 2)/2 = 2\frac{P(P+1)}{2}$. ⇒ Solve this and get P = 2. Now the sum of first 2 natural numbers is 1 + 2 = 3. The value of P + Q = 2 + 3 = 5 and its square i.e. 25 will be the answer.
3.	3	At the hundred's place 2 will come 49 times and 3 will come a single time. Thus the total at the hundred's place will be 2 × 49 + 3 = 101. At ten's place 5 will come nine times only but 6, 7, 8 and 9 each will come 10 times. So the total at ten's place = [5 × 9 + 10 × (6 + 7+ 8 + 9)] = 345. At unit's place all the ten digits including zero will come five times. So the total at the unit's place is 5 × (∑ 9) = 225. Value of G(300) is 101 + 345 + 225 = 671. So 3rd option is the answer. **OR** We can check the digital root of the series, (as discussed in the video of **DIGITAL ROOT**)
4.	4	*a* could be 3 or 5 or 17---------- So answer is infinite.
5.	2	Again this a unique property question, when the sum of four prime numbers is given to be odd, this means all the four prime numbers used are not odd i.e. at least one even prime number is used i.e. 2. So the product of four numbers must be even. Only 2nd option is the possible answer.
6.	4	The only possible value of K is 22 as it has to be even also. The sum of first K natural no. will be $\frac{22(22+1)}{2}$. Thus it can be seen that the sum is divisible by all the options given. Thus 4th option is the answer.
7.	4	It is particular feature about the two digit numbers, if you interchange the digits and if the difference between the digits is 1, the numbers (original and interchanged) will show a difference of 9. Similarly if the difference in the digits is 2, then the numbers will show a difference of 18 (27 if the difference is 3) and so on. Here it can be found from the question that the difference between the digits is 36 ÷ 9 = 4. But nothing can be said about the sum of the digits. So, answer is 4th option
8.	2	The difference is 4, thus 2nd option is the answer.
9.	4	In option 4, (x-y) is even, and any number multiplied by an even number is always even.
10.	3	See every term in the numerator is a multiple of 'n' except the last term. Now if this whole expression has to be an integer then even 'x' should be a multiple of 'n'. But the question is asking that it is not an integer that is possible when 'x' is not a multiple of 'n'. Hence 3rd option.